

into the product of two primes. Till now, this is non-trackable in the classical computer paradigm because it starts with a big enough integer — so it would take years to get the private key out, when compromised. Roughly, the key generation algorithm can be stated as below:

1. Get two equal sized random prime numbers, say p and q .
2. Make sure the product of p and q (let's call it n) is of the desired bit length, say 1024.
3. Let's compute the value of n and $\phi = (p-1)(q-1)$.
4. Let's select a proper integer e between 1 and ϕ .
5. Secret component derivation: d , again between 1 and ϕ , $ed = 1 \pmod{\phi}$.
6. The public key part is (n, e) and private part is (d, p, q) .
7. d, p, q and ϕ are secrets.

Quantum cyber security perspective

In this section we are going to discuss the proliferation of quantum computers in the cyber security space. The prime factorisation we discussed above, when tried out in classical computing, takes a very long time to be computed. This makes it practically invincible. We feel secure that our classical security model is intact and in place. However, that's not the case with the quantum computing paradigm. Figure 3 describes a mind-blowing comparison of prime factorisation across classical and quantum computing paradigms. The corresponding quantum computing prime factorisation algorithm is known as Shor's Algorithm.

Figure 3 vividly depicts the potential risks of the proliferation of quantum computing in the cyber security domain. To understand the contours of cyber security in the quantum age we need to first define what 'Quantum safe security' is.

Quantum safe security is often called 'post quantum security' as well.

FUTURE DANGER

Experts who think there is >50% chance of a quantum threat and the number of years in which it will hit.

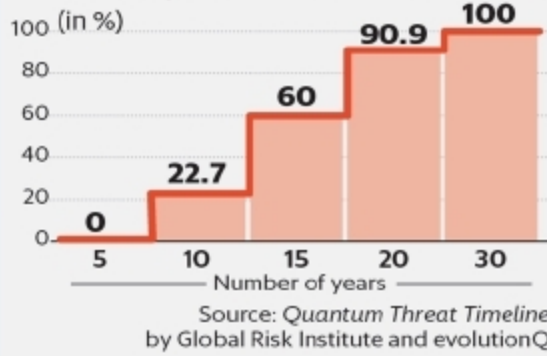


Figure 1: Quantum threat timeline

Just like the modelling of a security problem, let's start with the worst case scenario. Let's say the honest party has the classical computer and the adversary is quantum powered. Let's assume the adversary's quantum power does not necessarily exist today, since this can be a futuristic scenario also, to make the problem statement interesting. Now, in the classical world, we would like to build a security mechanism that cannot be compromised, even by the quantum powered adversary. So, winning in the unfavourable condition, is called post quantum security. This is one area attracting intense research interest today. Let's look at how to achieve such a post quantum security paradigm.

WORK IN PROGRESS

THE US passed the National Quantum Initiative Act in 2018 and plans to invest \$1.2 billion over the next five years.

CHINA IS investing \$10 billion to build the National Laboratory for Quantum Information Sciences.

THE EUROPEAN Commission will invest €1 billion to use quantum technology in commercial products.

PRIVATE INVESTMENT received by quantum tech firms in 2017 and 2018 totalled about \$450 million.

43% OF patents from quantum tech between 2012 and 2017 have come from Chinese firms and universities.

Source: Nature, European Commission's joint research centre in Italy, Mint research

Figure 2: Investments in quantum computing

To understand this seemingly unbelievable world of post quantum security, we need to understand what sort of problems a quantum computer can potentially solve. Can it solve any NP-complete problem? It cannot. Figure 4 depicts the BQP (Bounded Error Polynomial) time, which is the problem zone quantum computers can solve in polynomial time.

As we can see in this figure, there still exists an unsolvable zone which is called *quantum hard*. This means that it's beyond the capabilities of classical as well as quantum computers.

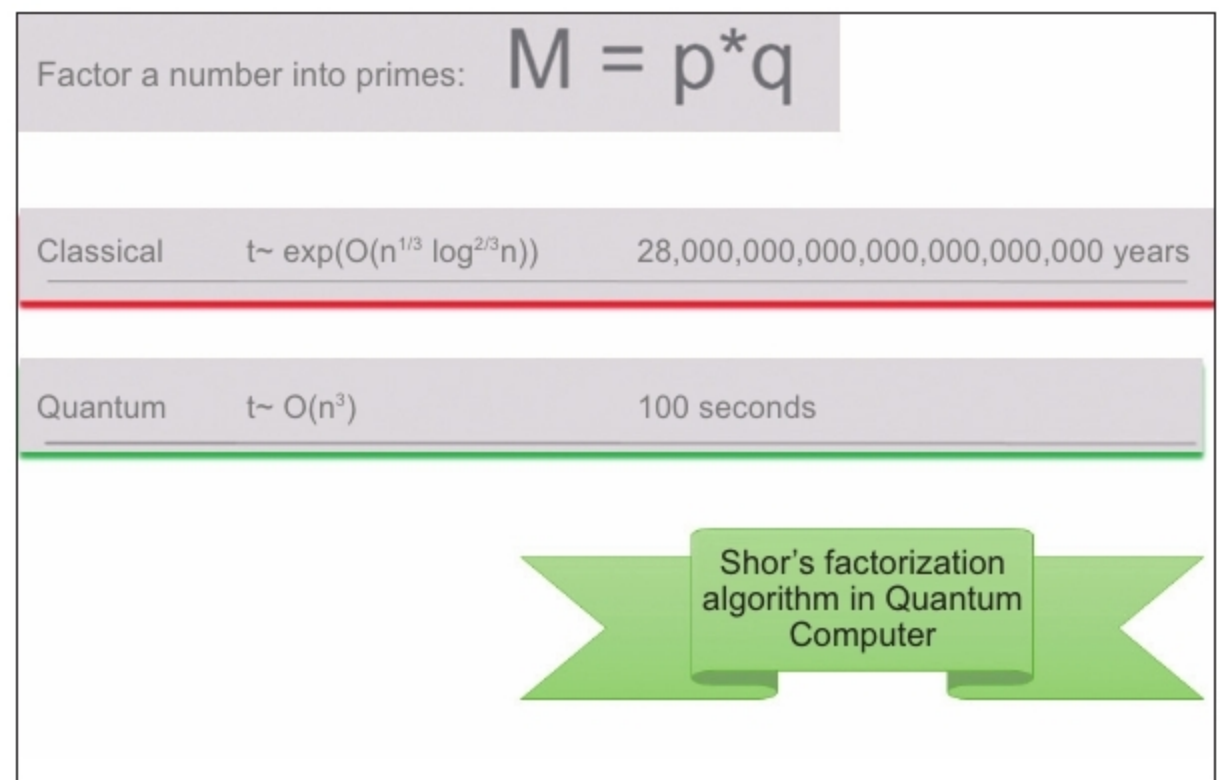


Figure 3: A comparison of computational time for prime factorisation in classical and quantum computing (Shor's algorithm)